

# Sales Data Analysis Using R

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## Sales Data Analysis Using R

### Introduction

This project analyzes sales data using R.

The dataset contains information about products, categories, quantity sold, unit price, sales amount, and regions.

R is used to organize the data, perform statistical analysis, create graphs, and understand sales patterns.

### Objectives

- Create sales data
- Check and clean the data
- Perform statistical analysis
- Analyze product sales
- Compare regional sales
- Create graphs
- Interpret the results
- Identify important findings
- Give a conclusion

### Project Background

Sales analysis helps businesses understand their sales performance.

By analyzing sales data, we can compare products, quantities, prices, and different regions.

In this project, R is used to perform basic data analysis and visualization.

### Scope of the Project

This project focuses on 20 sales records.

The main variables are:

- Date
- Product
- Category
- Quantity
- Unit Price
- Region
- Sales

# 1. Create Sales Data

We create a sales dataset using the `data.frame()` function.

```
sales <- data.frame(  
  Date = c(  
    "2026-01-05", "2026-01-08", "2026-01-12", "2026-01-18", "2026-01-23",  
    "2026-02-03", "2026-02-09", "2026-02-15", "2026-02-21", "2026-02-27",  
    "2026-03-04", "2026-03-10", "2026-03-16", "2026-03-22", "2026-03-28",  
    "2026-04-05", "2026-04-11", "2026-04-17", "2026-04-23", "2026-04-29"  
  ),  
  Product = c(  
    "Laptop", "Mouse", "Keyboard", "Monitor", "Headphones",  
    "Laptop", "Mouse", "Keyboard", "Monitor", "Headphones",  
    "Laptop", "Mouse", "Keyboard", "Monitor", "Headphones",  
    "Laptop", "Mouse", "Keyboard", "Monitor", "Headphones"  
  ),  
  Category = c(  
    "Electronics", "Accessories", "Accessories", "Electronics", "Accessories",  
    "Electronics", "Accessories", "Accessories", "Electronics", "Accessories",  
    "Electronics", "Accessories", "Accessories", "Electronics", "Accessories",  
    "Electronics", "Accessories", "Accessories", "Electronics", "Accessories"  
  ),  
  Quantity = c(  
    3, 15, 10, 6, 12,  
    4, 18, 12, 7, 15,  
    5, 20, 14, 8, 18,  
    6, 22, 16, 9, 20  
  ),  
  Unit_Price = c(  
    55000, 1200, 2200, 18000, 2500,  
    55000, 1200, 2200, 18000, 2500,  
    55000, 1200, 2200, 18000, 2500,  
    55000, 1200, 2200, 18000, 2500  
  ),  
  Region = c(  
    "North", "West", "South", "East", "West",  
    "South", "North", "East", "West", "South",  
    "East", "West", "North", "South", "East",  
    "West", "South", "East", "North", "West"  
  )  
)  
  
sales$Sales <- sales$Quantity * sales$Unit_Price  
  
sales
```

| ##    | Date       | Product    | Category    | Quantity | Unit_Price | Region | Sales  |
|-------|------------|------------|-------------|----------|------------|--------|--------|
| ## 1  | 2026-01-05 | Laptop     | Electronics | 3        | 55000      | North  | 165000 |
| ## 2  | 2026-01-08 | Mouse      | Accessories | 15       | 1200       | West   | 18000  |
| ## 3  | 2026-01-12 | Keyboard   | Accessories | 10       | 2200       | South  | 22000  |
| ## 4  | 2026-01-18 | Monitor    | Electronics | 6        | 18000      | East   | 108000 |
| ## 5  | 2026-01-23 | Headphones | Accessories | 12       | 2500       | West   | 30000  |
| ## 6  | 2026-02-03 | Laptop     | Electronics | 4        | 55000      | South  | 220000 |
| ## 7  | 2026-02-09 | Mouse      | Accessories | 18       | 1200       | North  | 21600  |
| ## 8  | 2026-02-15 | Keyboard   | Accessories | 12       | 2200       | East   | 26400  |
| ## 9  | 2026-02-21 | Monitor    | Electronics | 7        | 18000      | West   | 126000 |
| ## 10 | 2026-02-27 | Headphones | Accessories | 15       | 2500       | South  | 37500  |
| ## 11 | 2026-03-04 | Laptop     | Electronics | 5        | 55000      | East   | 275000 |
| ## 12 | 2026-03-10 | Mouse      | Accessories | 20       | 1200       | West   | 24000  |
| ## 13 | 2026-03-16 | Keyboard   | Accessories | 14       | 2200       | North  | 30800  |
| ## 14 | 2026-03-22 | Monitor    | Electronics | 8        | 18000      | South  | 144000 |
| ## 15 | 2026-03-28 | Headphones | Accessories | 18       | 2500       | East   | 45000  |
| ## 16 | 2026-04-05 | Laptop     | Electronics | 6        | 55000      | West   | 330000 |
| ## 17 | 2026-04-11 | Mouse      | Accessories | 22       | 1200       | South  | 26400  |
| ## 18 | 2026-04-17 | Keyboard   | Accessories | 16       | 2200       | East   | 35200  |
| ## 19 | 2026-04-23 | Monitor    | Electronics | 9        | 18000      | North  | 162000 |
| ## 20 | 2026-04-29 | Headphones | Accessories | 20       | 2500       | West   | 50000  |

## Dataset Description

The dataset contains 20 sales records.

The variables used in the dataset are:

- **Date:** Date of the sale
- **Product:** Name of the product
- **Category:** Product category
- **Quantity:** Number of units sold
- **Unit Price:** Price of one unit
- **Region:** Sales region
- **Sales:** Total sales amount

## 2. Check the Data

We use `summary()` to get a basic overview of the dataset.

```
summary(sales)
```

```
##           Date           Product           Category           Quantity           Unit_Price
## Length   :20   Length   :20   Length   :20   Min.    : 3.00   Min.    : 1200
## N.unique :20   N.unique : 5   N.unique : 2   1st Qu.: 6.75   1st Qu.: 2200
## N.blank  : 0   N.blank  : 0   N.blank  : 0   Median :12.00   Median : 2500
## Min.nchar:10   Min.nchar: 5   Min.nchar:11   Mean    :12.00   Mean    :15780
## Max.nchar:10   Max.nchar:10   Max.nchar:11   3rd Qu.:16.50   3rd Qu.:18000
##                                     Max.    :22.00   Max.    :55000
##           Region           Sales
## Length   :20   Min.    : 18000
## N.unique : 4   1st Qu.: 26400
## N.blank  : 0   Median : 41250
## Min.nchar: 4   Mean    : 94845
## Max.nchar: 5   3rd Qu.:148500
##                                     Max.    :330000
```

The summary provides information about the numerical variables in the dataset.

### 3. Check Dataset Structure

The `str()` function is used to understand the structure of the dataset.

```
str(sales)
```

```
## 'data.frame':   20 obs. of  7 variables:
## $ Date       : chr  "2026-01-05" "2026-01-08" "2026-01-12" "2026-01-18" ...
## $ Product    : chr  "Laptop" "Mouse" "Keyboard" "Monitor" ...
## $ Category   : chr  "Electronics" "Accessories" "Accessories" "Electronics" ...
## $ Quantity   : num  3 15 10 6 12 4 18 12 7 15 ...
## $ Unit_Price : num  55000 1200 2200 18000 2500 55000 1200 2200 18000 2500 ...
## $ Region     : chr  "North" "West" "South" "East" ...
## $ Sales      : num  165000 18000 22000 108000 30000 220000 21600 26400 126000 37500 ...
```

This helps us understand the number of observations, variables, and their data types.

### 4. Check Missing Values and Duplicates

We check whether the dataset contains missing values or duplicate records.

```
colSums(is.na(sales))
```

```
##           Date           Product           Category           Quantity           Unit_Price           Region           Sales
##           0             0             0             0             0             0             0
```

```
sum(duplicated(sales))
```

```
## [1] 0
```

The dataset contains no missing values and no duplicate records.

Therefore, the data is ready for analysis.

## 5. Basic Statistical Analysis

We calculate important statistics for quantity and sales.

```
mean(sales$Quantity)
```

```
## [1] 12
```

```
max(sales$Quantity)
```

```
## [1] 22
```

```
min(sales$Quantity)
```

```
## [1] 3
```

```
mean(sales$Sales)
```

```
## [1] 94845
```

```
max(sales$Sales)
```

```
## [1] 330000
```

```
min(sales$Sales)
```

```
## [1] 18000
```

These values help us understand the average, highest, and lowest sales performance.

## 6. Total Sales

We calculate the total sales from all records.

```
total_sales <- sum(sales$Sales)
```

```
total_sales
```

```
## [1] 1896900
```

The total sales represent the overall sales amount recorded in the dataset.

## 7. Sales by Product

We calculate total sales for each product.

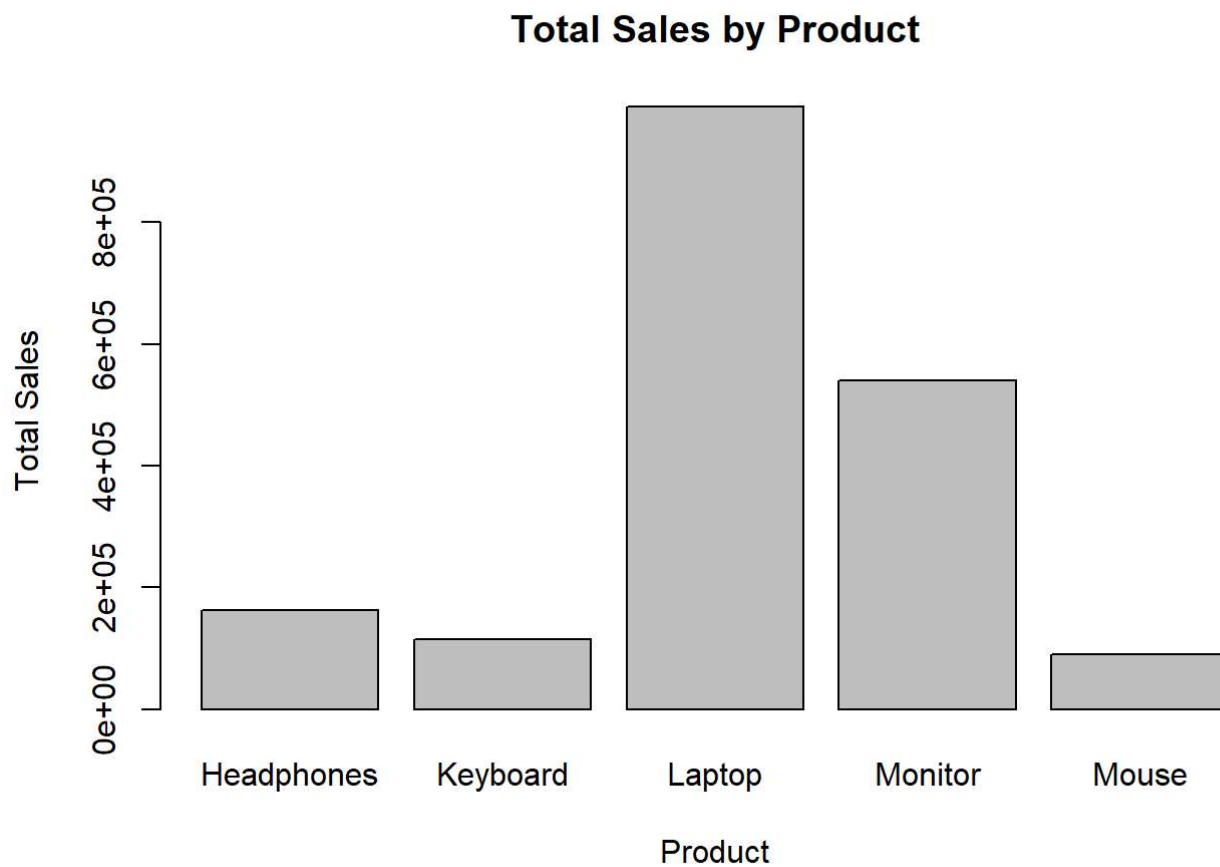
```
product_sales <- aggregate(  
  Sales ~ Product,  
  data = sales,  
  FUN = sum  
)  
  
product_sales
```

```
##      Product  Sales  
## 1 Headphones 162500  
## 2   Keyboard 114400  
## 3    Laptop 990000  
## 4   Monitor 540000  
## 5     Mouse  90000
```

This table helps compare the sales performance of different products.

## Product Sales Graph

```
barplot(  
  product_sales$Sales,  
  names.arg = product_sales$Product,  
  main = "Total Sales by Product",  
  xlab = "Product",  
  ylab = "Total Sales"  
)
```



The graph provides a visual comparison of total sales between different products.

## 8. Quantity Sold by Product

We calculate the total quantity sold for each product.

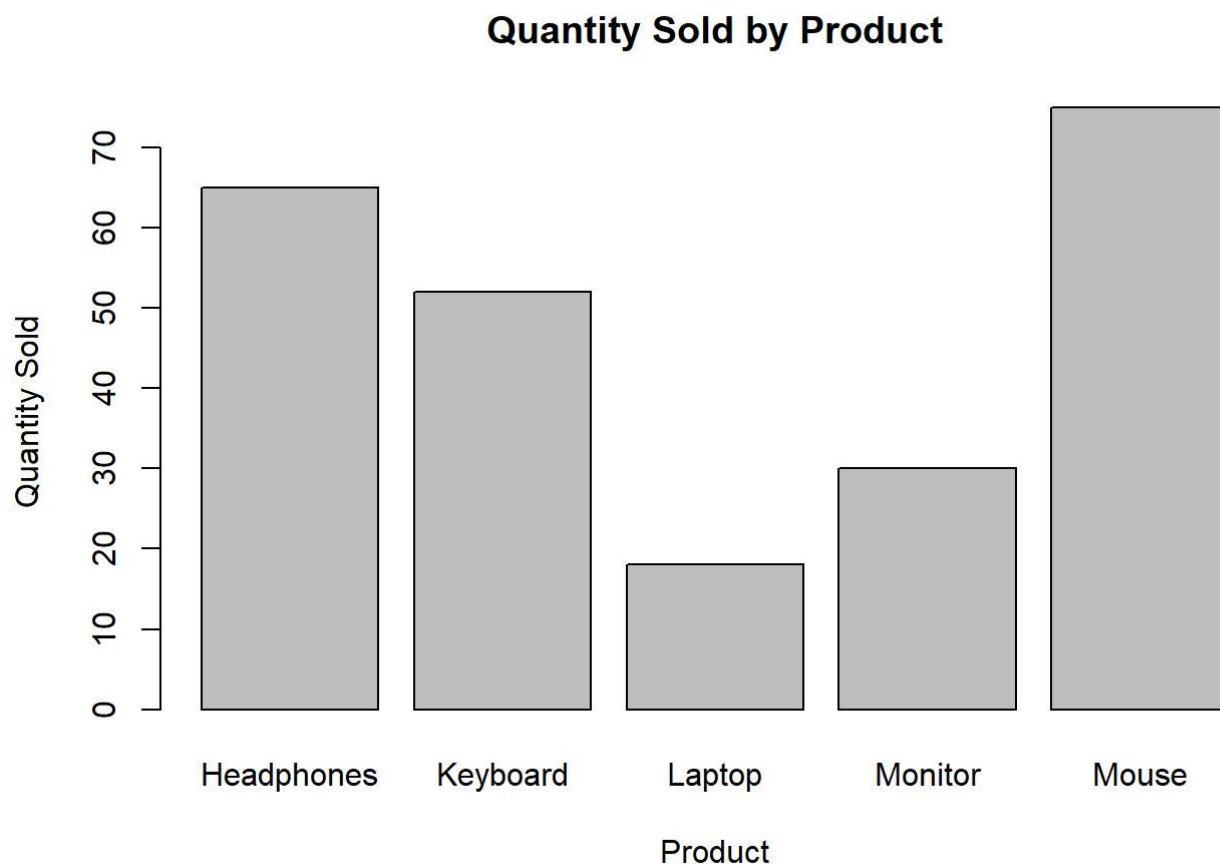
```
product_quantity <- aggregate(  
  Quantity ~ Product,  
  data = sales,  
  FUN = sum  
)
```

```
product_quantity
```

```
##      Product Quantity  
## 1 Headphones      65  
## 2  Keyboard      52  
## 3   Laptop      18  
## 4   Monitor      30  
## 5    Mouse      75
```

## Quantity Sold Graph

```
barplot(  
  product_quantity$Quantity,  
  names.arg = product_quantity$Product,  
  main = "Quantity Sold by Product",  
  xlab = "Product",  
  ylab = "Quantity Sold"  
)
```



The graph shows the number of units sold for each product.

## 9. Sales by Region

We calculate total sales for each region.

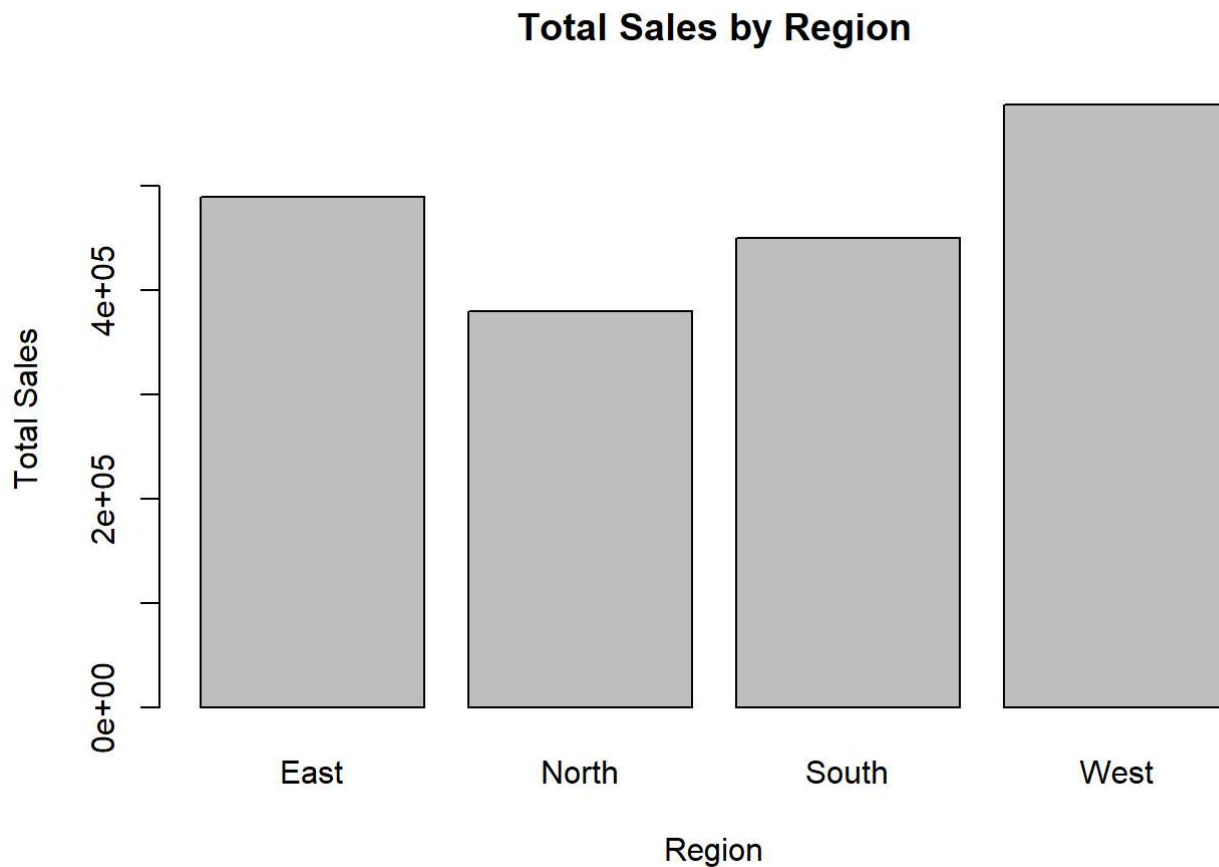
```
region_sales <- aggregate(  
  Sales ~ Region,  
  data = sales,  
  FUN = sum  
)  
  
region_sales
```



```
##   Region Sales
## 1   East 489600
## 2  North 379400
## 3  South 449900
## 4   West 578000
```

## Regional Sales Graph

```
barplot(
  region_sales$Sales,
  names.arg = region_sales$Region,
  main = "Total Sales by Region",
  xlab = "Region",
  ylab = "Total Sales"
)
```



The graph helps compare sales performance across regions.

## 10. Average Sales by Product

We calculate the average sales for each product.

```
average_product_sales <- aggregate(  
  Sales ~ Product,  
  data = sales,  
  FUN = mean  
)  
  
average_product_sales
```

```
##      Product  Sales  
## 1 Headphones 40625  
## 2   Keyboard 28600  
## 3    Laptop 247500  
## 4   Monitor 135000  
## 5     Mouse  22500
```

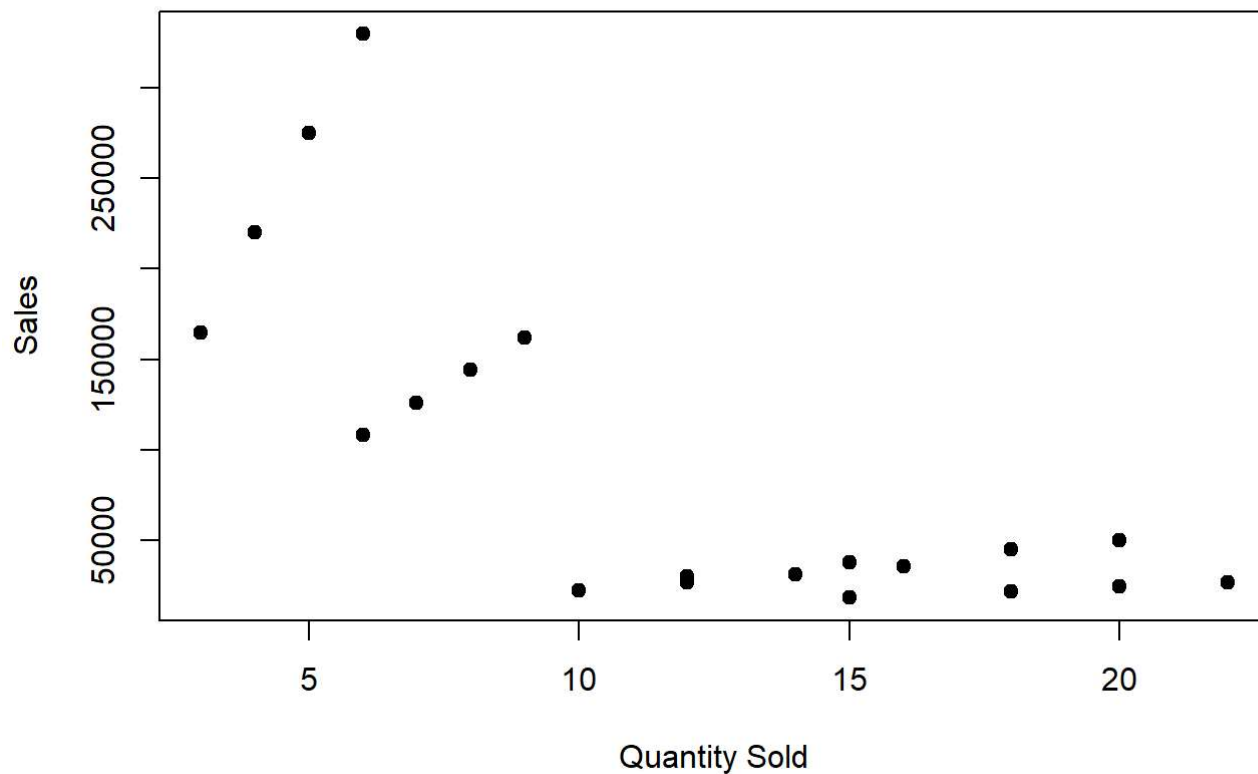
This analysis helps us understand the average sales generated by different products.

## 11. Quantity vs Sales

A scatter plot is used to understand the relationship between quantity sold and sales.

```
plot(  
  sales$Quantity,  
  sales$Sales,  
  main = "Quantity Sold vs Sales",  
  xlab = "Quantity Sold",  
  ylab = "Sales",  
  pch = 19  
)
```

## Quantity Sold vs Sales



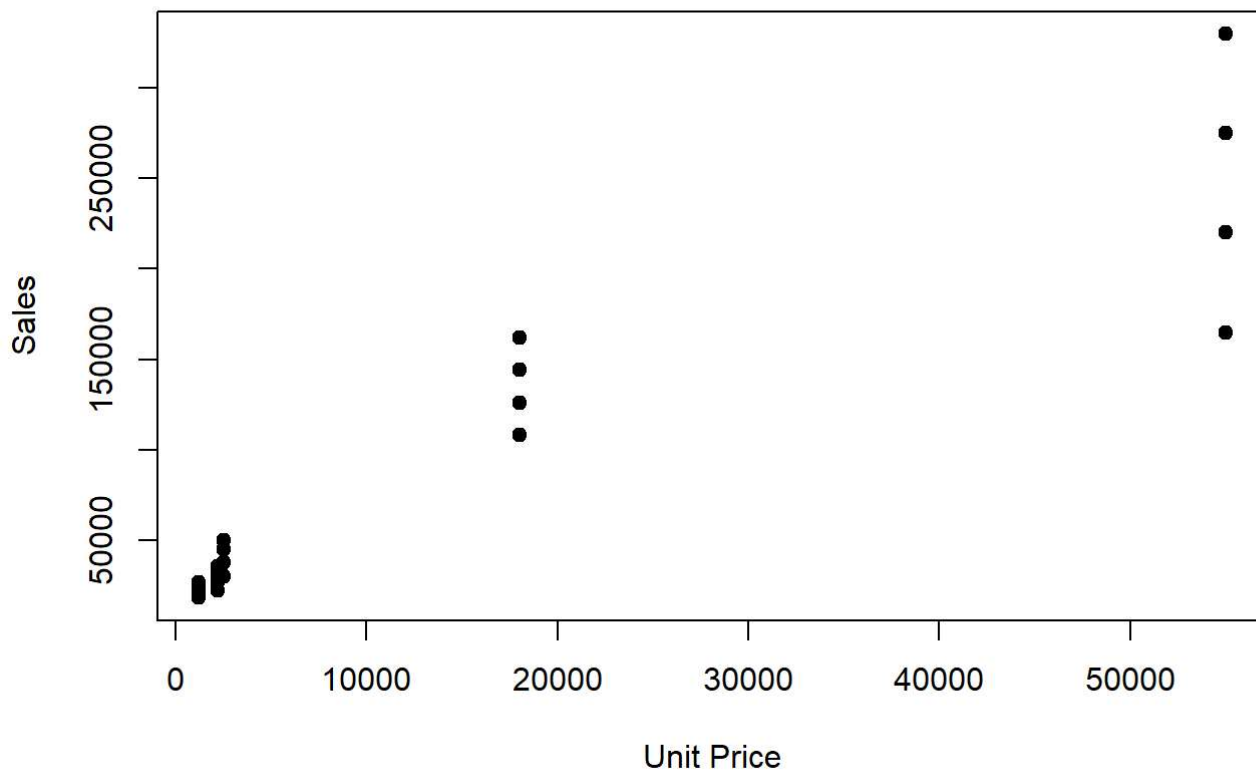
The graph shows the relationship between the number of units sold and the sales amount.

## 12. Unit Price vs Sales

We compare unit price with total sales.

```
plot(  
  sales$Unit_Price,  
  sales$Sales,  
  main = "Unit Price vs Sales",  
  xlab = "Unit Price",  
  ylab = "Sales",  
  pch = 19  
)
```

## Unit Price vs Sales



The graph helps us understand the relationship between unit price and sales.

## 13. Correlation Analysis

Correlation helps us understand the relationship between quantity sold and sales.

```
cor(sales$Quantity, sales$Sales)
```

```
## [1] -0.7544828
```

A positive value indicates a positive relationship between the two variables.

## 14. Highest Sales Record

We identify the record with the highest sales.

```
sales[which.max(sales$Sales), ]
```

```
##           Date Product    Category Quantity Unit_Price Region  Sales
## 16 2026-04-05  Laptop Electronics         6      55000   West 330000
```

This gives information about the product, quantity, price, region, and sales amount of the highest sales record.

## 15. Lowest Sales Record

We identify the record with the lowest sales.

```
sales[which.min(sales$Sales), ]
```

| ##   | Date       | Product | Category    | Quantity | Unit_Price | Region | Sales |
|------|------------|---------|-------------|----------|------------|--------|-------|
| ## 2 | 2026-01-08 | Mouse   | Accessories | 15       | 1200       | West   | 18000 |

This identifies the record with the lowest sales amount.

## 16. Interpretation of Results

The analysis provides several observations.

- Different products generate different sales amounts.
- Sales performance varies between regions.
- Quantity sold has a relationship with total sales.
- Product-wise analysis helps compare products.
- Regional analysis helps compare different regions.
- Graphs make the sales data easier to understand.

The results are based on the sample dataset used in this project.

## 17. Key Findings

The main findings are:

- Total sales can be calculated using `sum()`.
- Product sales can be compared using `aggregate()`.
- Regional sales can be compared using a bar graph.
- Quantity sold can be analyzed using statistical functions.
- Highest and lowest sales records can be identified.
- Graphs provide a visual representation of sales data.
- The dataset has no missing values.
- The dataset has no duplicate records.

## 18. Limitations

This project has some limitations.

- The dataset contains only 20 records.
- Only a limited number of products are included.
- Only four regions are considered.
- Customer information is not included.
- Profit and expenses are not included.
- The dataset is a small sample.

Therefore, the results should be interpreted according to the available data.

## 19. Future Scope

The project can be improved by using a larger dataset.

Additional information could include:

- Customer details
- Profit
- Discount
- Salesperson
- Monthly sales
- Customer ratings
- Payment method

Advanced statistical methods and machine learning could also be used for future sales prediction.

## 20. Conclusion

This project analyzed sales data using R.

The project demonstrated how to create a dataset, check its structure, identify missing values and duplicates, perform statistical analysis, and create graphs.

Product-wise and region-wise analysis helped us understand different aspects of sales performance.

Overall, R provides useful tools for data analysis, visualization, and interpretation of sales data.

## Final Summary

In this project, we learned how to:

- Create a dataset
- Check data structure
- Check missing values
- Check duplicate records
- Calculate statistics
- Calculate total sales
- Analyze products
- Analyze regions
- Create bar graphs
- Create scatter plots
- Calculate correlation
- Interpret results
- Identify findings
- Discuss limitations
- Suggest future improvements